

## Problem Set 6

### The Fundamental Theorem of Calculus

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#### Problem 3

(c) Both formulas remain the same.

#### Problem 4

(b)

$$\begin{aligned}\frac{d}{dx} \int_1^{x^4} \sec t \, dt \\ = \boxed{4x^3 \sec x^4}\end{aligned}$$

(c)

$$\begin{aligned}\frac{d}{dx} \int_x^\pi \sqrt{1 + \sec t} \, dt \\ = \boxed{-\sqrt{1 + \sec x}}\end{aligned}$$

(e)

$$\begin{aligned}\frac{d}{dx} \int_1^{\tan x} \sqrt{t + \sqrt{t}} \, dt \\ = \boxed{\sec^2 x \sqrt{\tan x + \sqrt{\tan x}}}\end{aligned}$$

(f)

$$\begin{aligned}\frac{d}{dx} \int_{1-3x}^1 \frac{v}{v^3 + 1} \, dv \\ = \boxed{\frac{3(1-3x)}{(1-3x)^3 + 1}}\end{aligned}$$

#### Problem 6

(a)

$$\begin{aligned}F'(x) &= \frac{d}{dx} \int_1^x f(t) \, dt = f(x) \\ F''(x) &= f'(x) = \frac{d}{dx} \int_1^{x^2} \frac{\sqrt{1+u^4}}{u} \, du = 2x \frac{\sqrt{1+x^8}}{x^2} \\ F''(2) &= 2 \cdot 2 \cdot \frac{\sqrt{1+2^8}}{4} = \boxed{\sqrt{257}}\end{aligned}$$

(b)

$$y' = \frac{d}{dx} \int_0^x \frac{1}{1+t+t^2} dt = \frac{1}{1+x+x^2}$$

$$y'' = \frac{d}{dx} \frac{1}{1+x+x^2} = -\frac{2x+1}{(1+x+x^2)^2}$$

$$y'' > 0 \implies 1+2x < 0 \implies \boxed{x \in (-\infty, -\frac{1}{2})}$$

(c)

$$6 + \int_a^x \frac{f(t)}{t^2} dt = 2\sqrt{x}$$

$$\frac{d}{dx} \left( 6 + \int_a^x \frac{f(t)}{t^2} dt \right) = \frac{d}{dx} 2\sqrt{x}$$

$$\frac{f(x)}{x^2} = \frac{1}{\sqrt{x}}$$

$$\boxed{f(x) = x\sqrt{x}}$$

$$6 + \int_a^x \frac{1}{\sqrt{x}} dt = 2\sqrt{x}$$

$$x = a \implies 6 + 0 = 2\sqrt{a}$$

$$\boxed{a = 9}$$

**Problem 7**

(a)

$$\int_1^2 h''(u) du = h'(2) - h'(1) = 5 - 2 = 3$$

The integral of  $h''(u)$  only depends on its anti-derivative, which is  $h'(u)$ .

(b)

$$\int_1^2 h'(u) du = h(2) - h(1) = 6 - (-2) = 8$$

(c)

$$V(20) - V(0) = \int_0^{20} V'(t) dt = \int_0^{20} r(t) dt$$

The initial value does not matter, as the answer only measures the net change in volume.

**Problem 8**

(a) As the function is positive for all values of  $x \neq 0$ , the area under the curve cannot possibly be negative. It fails because the function is not continuous on the interval.

**(b)**

(i)

$$\frac{g(x+h) - g(x)}{h} = \frac{1}{h} \left( \int_a^{x+h} f(t) dt - \int_a^x f(t) dt \right) = \frac{1}{h} \int_x^{x+h} f(t) dt$$

(ii)

$$\begin{aligned} m_h &\leq f(t) \leq M_h \\ \int_x^{x+h} m_h dt &\leq \int_x^{x+h} f(t) dt \leq \int_x^{x+h} M_h dt \\ h \cdot m_h &\leq g(x+h) - g(x) \leq h \cdot M_h \\ m_h &\leq \frac{g(x+h) - g(x)}{h} \leq M_h \end{aligned}$$

(iii)

$$\begin{aligned} \lim_{h \rightarrow 0^+} m_h &= \lim_{h \rightarrow 0^+} M_h = f(x) \\ \lim_{h \rightarrow 0^+} \frac{g(x+h) - g(x)}{h} &= \lim_{h \rightarrow 0^+} \frac{g(x+h) - g(x)}{h} = f(x) \end{aligned}$$

For  $h < 0$ , both the interval of integration and the value of  $h$  changes, but the result remains the same.

**(c)**

$$\begin{aligned} F'(x) &= g'(x) = f(x) \\ F(x) &= g(x) + C \\ F(a) &= g(a) + C = \int_a^a f(t) dt + C = C \\ F(b) &= g(b) + C = \int_a^b f(t) dt + C \\ \int_a^b f(t) dt &= F(b) - C = F(b) - F(a) \end{aligned}$$

**(d)**

(i)

$$\begin{aligned} x \in [-1, 0] &\implies g(x) = 0 \\ x \in (0, 1] &\implies g(x) = \int_0^x 1 dt = x \\ g(x) &= \begin{cases} 0, & x \in [-1, 0] \\ x, & x \in (0, 1] \end{cases} \end{aligned}$$

(ii)

$$\begin{aligned}g(0) &= \lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^+} g(x) = 0 \\ \lim_{x \rightarrow 0^-} g'(x) &= 0 \\ \lim_{x \rightarrow 0^+} g'(x) &= 1 \\ \lim_{x \rightarrow 0^-} g'(x) &\neq \lim_{x \rightarrow 0^+} g'(x)\end{aligned}$$

(iii) As  $f$  is not continuous on the interval, differentiability is not guaranteed for  $g(x)$ . The accumulation is good, as it renders  $g(x)$  a continuous function. However, sharp corners in the graph still exist.